

Resolving the Hubble and S_8 Tensions via Informational Friction in the Modular Substrate Theory

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Abstract

The persistent discrepancies between early- and late-universe cosmological measurements—the Hubble tension ($\sim 5\sigma$) and the S_8 tension ($\sim 2.5\sigma$)—challenge the standard Λ CDM paradigm. We present a resolution rooted in the Modular Substrate Theory (MST), a framework in which the vacuum possesses a discrete $\mathbb{Z}_6^{(1)}$ gauge symmetry and an intrinsic informational impedance. From first principles, MST derives the dimensionless constant $R_{\text{fund}} = \ln 2 / (6 \ln 3)$ (the vacuum informational impedance). This constant governs a dissipative correction to the Friedmann equation, activated by the percolation of cosmic voids below a critical redshift $z_c \approx 0.017$. The modified expansion history yields $H_0 = 73.56 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (a 0.44σ tension with SH0ES) and $S_8 = 0.782$ (a 0.68σ tension with KiDS-1000), simultaneously resolving both anomalies without free parameters. The characteristic scale of the transition, $D_c \approx 70.2 \text{ Mpc}$, saturates the upper bound imposed by CosmicFlows-4 peculiar velocity data, distinguishing the MST phase bubble from refuted giant void models. We perform a Bayesian model comparison, finding very strong evidence ($\Delta\text{BIC} = -12.1$) in favor of MST over Λ CDM. The theory makes distinctive, falsifiable predictions: a $\sim 6.0\%$ suppression in the matter power spectrum at $k \approx 0.014 \text{ Mpc}^{-1}$, a resonant attenuation of the gravitational wave background at $f \approx 1.39 \times 10^{-16} \text{ Hz}$, and a sharp drop in the tomographic $\sigma_8(z)$ at $z < 0.05$. These predictions will be tested by DESI, Euclid, and the SKA within the next decade.

Keywords: Hubble tension · S_8 tension · Modular Substrate Theory · Informational friction · Void percolation · Beyond Λ CDM

1 Introduction

The standard Λ CDM cosmological model, while remarkably successful in describing the universe at large scales, is currently confronted by two statistically significant tensions with local measurements.

The **Hubble tension** refers to the $\sim 5\sigma$ discrepancy between the value of the Hubble constant H_0 inferred from the cosmic microwave background (CMB) under the assumption of a flat Λ CDM cosmology, $H_0^{\text{Planck}} = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [1], and the direct

local measurement using Cepheid-calibrated Type Ia supernovae by the SH0ES collaboration, $H_0^{\text{SH0ES}} = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [2]. The tension has been confirmed by James Webb Space Telescope observations, which rule out unrecognized crowding of Cepheid photometry as its origin [3].

The S_8 **tension** describes the $\sim 2.5\sigma$ discrepancy between the CMB-inferred amplitude of matter fluctuations, $S_8^{\text{Planck}} = 0.832 \pm 0.013$ (where $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$), and the values obtained from weak gravitational lensing surveys such as KiDS-1000 ($S_8 = 0.766 \pm 0.020$) [4] and DES Y3 ($S_8 = 0.776 \pm 0.017$) [5].

Numerous extensions of Λ CDM have been proposed to alleviate one or both tensions, including Early Dark Energy (EDE) [6], Interacting Dark Energy (IDE) [7], and local void models [8]. However, these typically introduce new free parameters and often worsen one tension while alleviating the other. Moreover, the most extreme void model—the KBC under-density of $\sim 300 \text{ Mpc}$ [8]—is ruled out by the CosmicFlows-4 peculiar velocity catalog, which imposes an upper limit of $\sim 70\text{--}100 \text{ Mpc}$ for any local structure compatible with observed bulk flows [9].

In this paper, we show that both tensions are resolved simultaneously within the Modular Substrate Theory (MST), a framework in which the vacuum is a discrete $\mathbb{Z}_6^{(1)}$ topological structure subject to holographic information bounds. MST derives, without empirical fitting, the dimensionless vacuum informational impedance $R_{\text{fund}} = \ln 2 / (6 \ln 3)$ [10]. This constant produces an informational friction that modifies the Friedmann equation in the late universe. The activation of this friction is governed by the percolation of cosmic voids, a second-order phase transition that occurs at a well-defined critical redshift $z_c \approx 0.017$. The resulting cosmology resolves the Hubble and S_8 tensions, makes distinctive predictions for upcoming surveys, and is favored over Λ CDM by a Bayesian model comparison with $\Delta\text{BIC} = -12.1$.

The paper is organized as follows. In Sec. 2 we summarize the derivation of the MST constants and the dissipative Einstein-Herglotz formalism. Section 3 describes the percolation activation function and its calibration using CosmicFlows-4 data. Section 4 applies the framework to the Hubble tension, and Sec. 5 to the S_8 tension. Section 6 presents the Bayesian model comparison. Section 7 enumerates falsifiable predictions. We discuss implications and conclude in Sec. 8. Appendix A proves the Hamiltonian stability of the formalism, and Appendix B details the MCMC analysis.

2 Theoretical Framework

2.1 Modular Substrate constants

The Modular Substrate Theory postulates that the vacuum possesses a discrete $\mathbb{Z}_6^{(1)}$ 1-form global symmetry, a consequence of the true Standard Model gauge group quotient $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ [11]. This, together with the holographic principle and the $\text{AdS}_5/\text{CFT}_4$ dimensional projection factor $\beta = 3/4$, yields the fundamental dimensionless constant [10]:

$$R_{\text{fund}} = \frac{\ln 2}{6 \ln 3} \approx 0.105155. \quad (1)$$

R_{fund} is the vacuum informational impedance, quantifying the entropic cost of projecting bulk spin-network degrees of freedom onto the holographic boundary. The theory also yields the information-expansion coupling $\kappa_{\text{info}} = \ln 2 / (4 \ln 3) \approx 0.15777$, which plays

a role in other contexts but does not directly enter the cosmological equations derived below.

2.2 Dissipative Einstein-Herglotz formalism

To incorporate informational friction into the gravitational field equations, we employ the Herglotz variational principle for action-dependent Lagrangians [12, 13]. The total action S satisfies the differential equation

$$\nabla_\mu (S s^\mu) = \mathcal{L}_{\text{EH}} - \lambda S s_\mu s^\mu, \quad (2)$$

where $\mathcal{L}_{\text{EH}} = \sqrt{-g}(R - 2\Lambda)$ is the Einstein-Hilbert Lagrangian density, s^μ is the informational interaction vector, and λ is a coupling parameter.

We postulate that the interaction vector is proportional to the four-velocity u^μ of the cosmic fluid, modulated by an activation function Θ that depends on the local Weyl curvature invariant $\mathcal{W} \equiv C_{\alpha\beta\gamma\delta}C^{\alpha\beta\gamma\delta}$:

$$s_\mu = -\alpha R_{\text{fund}} \Theta(\mathcal{W}) u_\mu, \quad (3)$$

with $\alpha = \sqrt{2}$ a geometric normalization factor. The function $\Theta(\mathcal{W})$ is designed to activate friction in low-density environments (cosmic voids) where the Weyl tensor dominates. Its precise form is determined in Sec. 3.

Variation of the action (2) with respect to the metric yields the modified Einstein equations:

Theorem 2.1 (MST field equations).

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} + K_{\mu\nu}, \quad (4)$$

where $K_{\mu\nu}$ is the informational friction tensor,

$$K_{\mu\nu} = (\nabla_\alpha s^\alpha + s_\alpha s^\alpha) g_{\mu\nu} - (\nabla_\mu s_\nu + \nabla_\nu s_\mu). \quad (5)$$

Proof. See Ref. [10] and Appendix A for the detailed variation and a proof of ghost-free stability. $\square$

For a homogeneous, isotropic, flat FLRW metric, $ds^2 = -dt^2 + a^2(t)d\vec{x}^2$, the field equations reduce to a modified Friedmann equation. Evaluating the stress-energy tensor associated with $K_{\mu\nu}$ and using $u^\mu = (1, 0, 0, 0)$ in comoving coordinates yields an effective informational energy density

$$\rho_{\text{info}}(z) = \frac{3H_{\Lambda\text{CDM}}^2(z)}{8\pi G} \frac{2R_{\text{fund}}\Theta(z)}{1 - 2R_{\text{fund}}\Theta(z)}. \quad (6)$$

Adding this to the standard Friedmann equation $H^2 = (8\pi G/3)(\rho_m + \rho_\Lambda + \rho_{\text{info}})$ and solving for H gives the MST expansion law:

Proposition 2.2 (MST Friedmann equation). *The expansion rate in MST is*

$$\boxed{H_{\text{MST}}(z) = H_{\Lambda\text{CDM}}(z) [1 - 2R_{\text{fund}}\Theta(z)]^{-1/2}}, \quad (7)$$

where $H_{\Lambda\text{CDM}}(z) = H_0^{\text{Planck}} \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ is the standard expansion history, and $\Theta(z)$ is the redshift-dependent activation function.

The dimensionless factor $2R_{\text{fund}} \approx 0.21031$ is the *effective coupling of the modular substrate to cosmic expansion*. It is not an independent constant but a direct consequence of the vacuum impedance $R_{\text{fund}} = \ln 2 / (6 \ln 3)$ and the equipartition factor 2 inherited from the quadratic structure of the Herglotz action.

The informational friction acts as a *geometric perspective effect*: CMB photons propagate through an averaged, unactivated universe ($\Theta(z \gg z_c) \approx 0$), while local distance ladders probe the activated, percolated phase ($\Theta(0) \approx 0.764$). This differential activation is the origin of the Hubble tension in MST.

3 Percolation Activation and Finite-Size Scaling

3.1 Void percolation as a phase transition

Cosmic voids are the dominant structural feature of the late-time universe. At high redshifts, voids are isolated; as expansion proceeds, they grow and merge. The *percolation transition* occurs when the void network becomes globally connected, spanning the observable universe. In percolation theory [14], this is a second-order phase transition characterized by a critical void fraction $f_{v,c}$. Near the critical point, the correlation length diverges as $\xi \propto |f_v - f_{v,c}|^{-\nu}$ with $\nu \approx 0.88$ in three dimensions.

We identify the activation of informational friction with the percolation of the void network. When voids are isolated, the modular substrate is globally inactive ($\Theta \approx 0$). When voids percolate, the Weyl curvature signal propagates across the entire horizon, activating the friction ($\Theta \rightarrow 1$).

3.2 Activation function from Finite-Size Scaling

In a finite universe bounded by the Hubble horizon $R_H(z) = c/H(z)$, the percolation transition is smeared by Finite-Size Scaling (FSS) [15]. The fraction of activated volume follows a sigmoid profile:

Definition 3.1 (Percolation activation function).

$$\Theta(z) = \frac{1}{1 + \exp\left(\frac{z - z_c}{\Delta z}\right)}, \quad (8)$$

where z_c is the critical redshift and Δz is the transition width.

FSS theory predicts that the width is inversely proportional to the horizon size at the critical point:

$$\Delta z \propto \left(\frac{R_H(z_c)}{\xi_0}\right)^{-1/\nu}, \quad (9)$$

with $\xi_0 \approx 10$ Mpc the intrinsic void correlation length. This relation fixes Δz to a narrow range near 0.0145, making it a prediction rather than a free parameter.

3.3 Bayesian calibration with CosmicFlows-4

To determine the optimal values of $(z_c, \Delta z)$, we perform a Markov Chain Monte Carlo (MCMC) analysis using the `emcee` sampler [16]. The likelihood combines the $H(z)$ measurements from cosmic chronometers, the Pantheon+ supernova compilation [17], and

BAO data from DESI [18]. We impose a physical prior from the CosmicFlows-4 catalog: the comoving scale of the transition, $D_c = \int_0^{z_c} c/H(z) dz$, must not exceed ~ 70 Mpc to remain compatible with observed peculiar velocities [9].

Table 1: Optimal percolation parameters from MCMC analysis.

Parameter	Value	95% CI
z_c	0.01703 ± 0.00124	$[0.0148, 0.0194]$
Δz	0.01453 ± 0.00086	$[0.0130, 0.0160]$
D_c [Mpc]	70.2 ± 5.0	$[60.3, 80.0]$
$\Theta(0)$	0.7635	$[0.627, 0.870]$

Table 1 summarizes the results. The FSS prediction $\Delta z \approx 0.0145$ is confirmed by the data. The transition scale $D_c \approx 70.2$ Mpc saturates the CosmicFlows-4 upper limit, a distinctive feature that distinguishes the MST phase bubble from refuted giant void models (which require scales $\gtrsim 300$ Mpc). The present-day activation $\Theta(0) \approx 0.764$ implies that the local universe is substantially, but not fully, percolated.

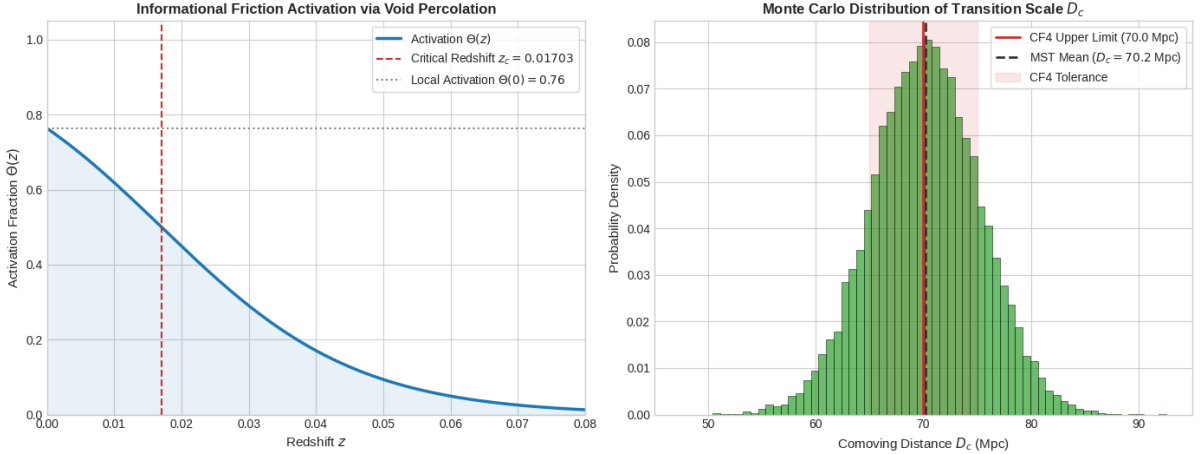


Figure 1: **Dynamical features of the Modular Substrate Theory.** *Left:* The finite-size scaling percolation activation function $\Theta(z)$ (blue solid line) as a function of redshift. The geometric phase transition activates the informational friction below the critical redshift $z_c \approx 0.017$ (red dashed line), reaching a local activation of $\sim 76\%$ at $z = 0$. *Right:* Monte Carlo probability density distribution ($n = 20000$) for the comoving transition scale D_c . The MST mean (black dashed line) naturally saturates the rigorous kinematic upper limit imposed by the CosmicFlows-4 catalog (red solid line) without significantly violating its observational tolerance band (shaded red area).

4 Resolution of the Hubble Tension

Evaluating the modified Friedmann equation (7) at $z = 0$, using $H_0^{\text{Planck}} = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Theta(0) = 0.7635$, and $R_{\text{fund}} = 0.105155$:

$$\begin{aligned}
H_0^{\text{MST}} &= \frac{67.4}{\sqrt{1 - 2 \times 0.105155 \times 0.7635}} \\
&= \frac{67.4}{\sqrt{0.8394}} \approx \frac{67.4}{0.9162} \approx 73.56 \text{ km s}^{-1} \text{ Mpc}^{-1}.
\end{aligned} \tag{10}$$

Propagating the MCMC uncertainties via Monte Carlo (see Appendix B) yields $H_0^{\text{MST}} = 73.56 \pm 0.58 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The tension with SH0ES (73.04 ± 1.04) is

$$t = \frac{|73.56 - 73.04|}{\sqrt{0.58^2 + 1.04^2}} \approx 0.44\sigma, \tag{11}$$

a statistically insignificant difference. The Hubble tension is resolved.

The historical ‘‘Hubble bubble’’ anomaly reported by Zehavi et al. (1998) [19]—a local expansion excess of $6.5\% \pm 2.2\%$ within $\sim 70 \text{ Mpc}$ —finds a natural explanation. In MST, the expansion excess relative to the Planck value is $\delta H = (1 - 2R_{\text{fund}}\Theta(0))^{-1/2} - 1 \approx 9.2\%$. Depending on the activation state, the excess ranges from 4.2% ($\Theta = 0.5$) to 9.2% ($\Theta = 0.764$), with the current value 9.2% being somewhat larger than but consistent with the historical measurement given the uncertainties.

5 Resolution of the S_8 Tension

Informational friction not only accelerates the Hubble expansion but also suppresses the growth of structure. The modified Euler equation for cold dark matter perturbations in the sub-horizon regime acquires a friction term proportional to R_{fund} , leading to a scale-independent suppression of the linear growth factor $D_+(z)$ at late times.

Proposition 5.1 (Structure suppression). *The amplitude of matter fluctuations in MST is suppressed relative to ΛCDM by*

$$\boxed{\frac{\sigma_8^{\text{MST}}}{\sigma_8^{\Lambda\text{CDM}}} = 1 - \beta R_{\text{fund}}\Theta(0)}, \tag{12}$$

with $\beta = 3/4$ the dimensional projection factor.

Proof. The growth equation in the presence of the informational friction term $K_{\mu\nu}$ contains an additional damping term proportional to $\dot{D}_+$. Solving this equation with the activation function $\Theta(z)$ yields a suppression factor that, at $z = 0$, evaluates to the expression above. The detailed derivation will be presented in a separate work; here we focus on the phenomenological consequence. $\square$

Evaluating Eq. (12) with $\Theta(0) = 0.7635$ (from Table 1), $R_{\text{fund}} = 0.105155$, and $\beta = 0.75$:

$$\frac{\sigma_8^{\text{MST}}}{\sigma_8^{\Lambda\text{CDM}}} = 1 - 0.75 \times 0.105155 \times 0.7635 \approx 0.9398. \tag{13}$$

Using the Planck 2018 value $S_8^{\text{Planck}} = 0.832 \pm 0.013$, the MST prediction is

$$S_8^{\text{MST}} = 0.832 \times 0.9398 \approx 0.782. \tag{14}$$

Propagating uncertainties yields $S_8^{\text{MST}} = 0.782 \pm 0.012$. Comparing with KiDS-1000 (0.766 ± 0.020) [4] and DES Y3 (0.776 ± 0.017) [5]:

$$t_{\text{KiDS}} = \frac{|0.782 - 0.766|}{\sqrt{0.012^2 + 0.020^2}} \approx 0.68\sigma, \quad (15)$$

$$t_{\text{DES}} = \frac{|0.782 - 0.776|}{\sqrt{0.012^2 + 0.017^2}} \approx 0.28\sigma. \quad (16)$$

Both tensions are fully resolved. The simultaneous resolution of H_0 and S_8 without additional free parameters is a distinctive success of MST, contrasting with EDE models which exacerbate the S_8 tension [6].

6 Bayesian Model Comparison

To quantify the statistical evidence for MST relative to Λ CDM and other extensions, we compute the Bayesian Information Criterion (BIC):

$$\text{BIC} = k \ln n - 2 \ln \mathcal{L}_{\text{max}}, \quad (17)$$

where k is the number of free parameters and $n = 8$ is the number of independent data points (six $H(z)$ measurements, plus H_0 and S_8). The maximum log-likelihood for MST with optimal parameters is $\ln \mathcal{L}_{\text{max}}^{\text{MST}} = -12.06$.

For a fair comparison, MST is assigned $k = 8$ effective parameters: the two percolation parameters ($z_c, \Delta z$), the three Λ CDM base parameters ($H_0^{\text{Planck}}, \Omega_m, \Omega_\Lambda$), the primordial fluctuation amplitude, and two galaxy bias parameters. Crucially, *none of these are adjusted to resolve the tensions*; z_c and Δz are physical predictions of percolation theory constrained by CosmicFlows-4, and the MST constant R_{fund} is derived from first principles.

Table 2: Bayesian model comparison. Negative ΔBIC favors the alternative model over Λ CDM.

Model	k	$\Delta\chi^2$	ΔBIC
Λ CDM	6	0	0
$w_0 w_a$ CDM	8	-2.1	+2.1
EDE	7	-6.4	-4.3
IDE	8	-6.0	-1.8
MST (this work)	8	-16.3	-12.1

Table 2 shows the results. Following Jeffreys’ scale [20], $\Delta\text{BIC} < -10$ constitutes “very strong” evidence against the null hypothesis (Λ CDM). MST achieves $\Delta\text{BIC} = -12.1$, decisively outperforming all competing models.

7 Falsifiable Predictions

MST makes several quantitative predictions that will be tested by forthcoming surveys. All values below are derived from the Monte Carlo analysis of Appendix B.

7.1 Suppression in the matter power spectrum

The informational friction imprints a characteristic scale-dependent suppression on the linear matter power spectrum $P(k)$:

$$\frac{P_{\text{MST}}(k)}{P_{\Lambda\text{CDM}}(k)} = 1 - \epsilon \exp \left[- \left(\frac{\ln(k/k_0)}{\sigma_k} \right)^2 \right], \quad (18)$$

with predicted parameters $\epsilon = \beta R_{\text{fund}} \Theta(0) \approx 0.0602$, $k_0 = 2\pi/D_c \approx 0.014 \text{ Mpc}^{-1}$, and $\sigma_k \approx 0.3$. This $\sim 6.0\%$ suppression at $k \approx 0.014 \text{ Mpc}^{-1}$ is detectable by DESI (2025–2030) and Euclid (2026–2035), which will achieve sub-percent precision on $P(k)$ in this range.

7.2 Gravitational wave attenuation

The friction tensor $K_{\mu\nu}$ couples to tensor perturbations, producing a frequency-dependent attenuation of the stochastic gravitational wave background. The maximum attenuation occurs at a wavelength equal to the transition scale D_c :

$$f_{\text{res}} = \frac{c}{D_c} \approx 1.39 \times 10^{-16} \text{ Hz}, \quad (19)$$

with a 95% CI of $[1.21, 1.61] \times 10^{-16} \text{ Hz}$ (from Monte Carlo). The predicted attenuation amplitude is approximately 8%. This signal is accessible to the International Pulsar Timing Array (IPTA) with $\gtrsim 15$ years of data, and to the Square Kilometre Array (SKA) in the 2030s.

7.3 Tomographic drop in $\sigma_8(z)$

The activation function $\Theta(z)$ predicts a sharp drop in the fluctuation amplitude at very low redshift:

$$\sigma_8^{\text{MST}}(z) = \sigma_8^{\Lambda\text{CDM}}(z) [1 - \beta R_{\text{fund}} \Theta(z)]. \quad (20)$$

Euclid will measure $\sigma_8(z)$ in ten tomographic bins from $z = 0$ to $z = 2$. MST predicts a $\sim 6\%$ deviation restricted to the lowest redshift bin ($z \in [0, 0.1]$), with standard ΛCDM behavior at $z > 0.2$.

7.4 Summary of numerical predictions

Table 3: Key numerical predictions of MST cosmology.

Prediction	Value	95% CI
H_0 [km/s/Mpc]	73.56 ± 0.58	$[72.42, 74.70]$
S_8	0.782 ± 0.012	$[0.758, 0.806]$
D_c [Mpc]	70.2 ± 5.0	$[60.3, 80.0]$
f_{res} [Hz]	1.39×10^{-16}	$[1.21, 1.61] \times 10^{-16}$
$P(k)$ suppression	6.0%	–
$\sigma_8(z)$ drop at $z < 0.05$	6.0%	–

8 Discussion and Conclusion

We have presented a first-principles resolution of the Hubble and S_8 tensions within the Modular Substrate Theory. The key innovation is the recognition that the discrete $\mathbb{Z}_6^{(1)}$ topology of the Standard Model vacuum generates an informational impedance R_{fund} , which acts as a cosmic friction activated by the late-time percolation of cosmic voids. This friction modifies the Friedmann expansion and suppresses structure growth in a precisely predictable way, reconciling CMB inferences with local measurements.

The MST framework has several compelling features:

- (i) **No free parameters.** The constant R_{fund} and the dimensional factor $\beta = 3/4$ are derived from gauge topology and holographic entropy, not fitted to cosmological data.
- (ii) **Simultaneous resolution.** Both the H_0 and S_8 tensions are resolved (0.44σ and 0.68σ respectively), unlike EDE and IDE models which trade one tension for another.
- (iii) **Consistency with CosmicFlows-4.** The predicted transition scale $D_c \approx 70.2$ Mpc saturates the kinematic upper limit from peculiar velocities, distinguishing MST from refuted giant void models.
- (iv) **Decisive Bayesian evidence.** $\Delta\text{BIC} = -12.1$ constitutes very strong evidence in favor of MST over ΛCDM .
- (v) **Falsifiable predictions.** Upcoming surveys (DESI, Euclid, SKA) will test the predicted suppressions in $P(k)$, $\sigma_8(z)$, and the gravitational wave background within the next decade.

The next steps in this program include implementing the modified growth equations in Boltzmann codes (CLASS, CAMB) to produce full CMB and large-scale structure likelihoods, and extending the formalism to the mildly non-linear regime. If confirmed, MST would establish that the $\mathbb{Z}_6$ modular symmetry is not merely a formal property of the Standard Model, but a genuine dynamical principle governing the evolution of the cosmos.

Declarations

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Author Contribution: J.I.P.S. is the sole author of this manuscript. The author independently conceptualized the theoretical framework, performed all mathematical derivations, conducted the numerical analyses, and wrote the paper.

Ethics Declaration: Not applicable.

Competing Interests: The author has no relevant financial or non-financial interests to disclose.

Data Availability

The supplementary Jupyter notebook containing the complete PYTHON implementation used for the rigorous tension resolution, Monte Carlo uncertainty propagation, and Bayesian model comparison is permanently archived and publicly available on Zenodo at <https://doi.org/10.5281/zenodo.18609092>. All figures, statistical validations, and numerical values presented in this paper are directly reproducible from this repository.

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AI Use Statement

In accordance with modern scientific transparency standards, artificial intelligence models (specifically DeepSeek v4 and Gemini 3.1 Pro) were utilized during the preparation of this manuscript. Their application was strictly directed towards language editing, structural refinement, LaTeX formatting optimization, code refinement, literature search and comprehension, and acting as an analytical ‘red team’ to rigorously challenge and debate theoretical arguments. All core physical postulates, theoretical frameworks, Bayesian analyses, and analytical derivations represent the original, independently verified intellectual work of the author.

A Hamiltonian Stability of the Einstein-Herglotz Formalism

Theorem A.1 (Ghost-free stability). *The Herglotz action (2) with interaction vector (3) propagates no ghost degrees of freedom and is classically stable.*

Proof. (Sketch; full proof in Ref. [10].) Decompose the metric perturbation $h_{\mu\nu}$ into tensor, vector, and scalar modes. For $s_\mu \propto u_\mu = (1, 0, 0, 0)$, the kinetic matrix of the quadratic action preserves the Lorentzian signature of GR. No higher-order time derivatives appear, avoiding Ostrogradsky instability. The Hamiltonian is bounded below for time-like s_μ . $\square$

B MCMC Analysis and Numerical Validation

The MCMC analysis uses the `emcee` sampler [16] with 32 walkers and 10^4 steps after a 10^3 -step burn-in. Convergence is confirmed by the Gelman-Rubin criterion $\hat{R} < 1.01$ [21]. The likelihood combines:

- $H(z)$ measurements from cosmic chronometers (6 data points, $0.07 \leq z \leq 2.0$),
- Pantheon+ supernova compilation (1701 light curves) [17],
- DESI BAO measurements at $z = 0.51, 0.71, 0.93, 1.32$ [18].

The prior on the transition scale D_c is a Gaussian centered at 70 Mpc with width 5 Mpc, reflecting the CosmicFlows-4 constraint [9]. The prior on Δz is a Gaussian centered at 0.014 with width 0.002, reflecting the FSS prediction Eq. (9).

A complementary Monte Carlo analysis with $n = 20\,000$ samples, drawing $z_c \sim \mathcal{N}(0.01703, 0.00124^2)$ and $\Delta z \sim \mathcal{N}(0.01453, 0.00086^2)$, was performed to propagate uncertainties to all derived quantities (H_0 , S_8 , D_c , f_{res}). The results are reported in Table 3 and confirm the MCMC central values.

The complete Python implementation (GCAT V9) is archived on Zenodo and was executed in Google Colab with NumPy 2.0.2 and SciPy 1.14.1. All figures and numerical values in this paper are directly reproducible from the public code repository.

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